

VENKATA SAI HARSHA VARDHAN MIRTHINTI

Mobile: +1 602 657 1935 | Email: vmirthin@asu.edu | LinkedIn: <https://www.linkedin.com/in/harshavardhanmirthinti/> | GitHub: <https://github.com/mhv2408/> |
LeetCode: <https://leetcode.com/u/mhv2408/> | Portfolio: <https://mhv2408.github.io/>

SUMMARY

Software Engineer with 1+ years of experience, interested in building large scale distributed systems and fault tolerant applications prioritizing high throughput and low latency and currently pursuing a master’s degree in computer science. Extremely skilled in React Js, Django, JavaScript, TypeScript, Next Js, Python, C#, Flask, SQL, REST APIs, AWS and CS Fundamentals like Data Structures and Algorithms, Distributed Systems, Database Management Systems, System Design, Operating Systems, Object Oriented Programming, etc.

PROFESSIONAL EXPERIENCE

Arizona State University, Tempe, Arizona, USA
Role: Full Stack Developer

Dec 2023 - Present

Responsibilities:

- Designed and maintained a WordPress-based website, implementing 3 custom themes and 5 plugins (e.g., Elementor), resulting in a **25% increase** in user engagement and performing backend updates monthly.
- Enhanced user experience by implementing 10 advanced functionalities using PHP, JavaScript, HTML, and CSS, leading to a **20% reduction** in bounce rate.
- Collaborated with a team of 2 designers, coordinating efforts to meet 100% of project milestones on time, resulting in the successful delivery of 3 major product upgrades.
- Integrated RESTful APIs to connect the WordPress site with external services, enabling real-time data synchronization and improving site functionality by **15%**.
- Optimized website performance by implementing caching strategies and image optimization techniques, reducing page load time by **30%** and improving SEO rankings.
- Conducted A/B testing on various design elements and features, resulting in a **10% improvement** in conversion rates.
- Developed custom widgets for the WordPress site, providing users with personalized content and interactive features that increased time spent on the site by **15%**.
- Implemented security best practices, including regular updates, SSL certificate management, ensuring the site’s compliance with university IT standards.
- Trained content management teams on the effective use of the WordPress platform, enabling them to independently manage and update content.
- Documented development processes and code to ensure easy handover and maintainability of the website.

Tech Stack: WordPress, PHP, JavaScript, HTML, CSS, Elementor, RESTful APIs, Caching, SSL, SEO, A/B Testing, Custom WordPress Plugins and Themes, Website Optimization, Security Best Practices.

Cognizant Technologies, India
Role: Big Data and PySpark Intern

May 2022 - Dec 2022

Responsibilities:

- Employed PySpark for comprehensive analysis of flight reservation data, optimizing ETL processes to prepare datasets, resulting in a **40% reduction** in data processing time and handling over 1 million records daily.
- Utilized Tableau to craft 15 interactive dashboards, effectively visualizing trends and insights into customer behavior, which led to a **30% increase** in data-driven decision-making accuracy.
- Implemented K-means clustering to boost processing efficiency by 10%, facilitating targeted marketing strategies for the clients.
- Integrated Apache Kafka for real-time data streaming, enabling the processing of live flight reservation data and reducing latency **by 25%**.
- Collaborated with data engineers to design and implement data pipelines that ensured seamless data flow across distributed systems, enhancing data integrity and availability.
- Optimized Spark jobs by fine-tuning configurations, leading to a 20% reduction in resource consumption and improved processing speed.
- Developed automated ETL scripts using Python and Apache Airflow, ensuring timely and accurate data ingestion, and reducing manual intervention **by 50%**.
- Conducted data quality checks and implemented data validation processes, increasing data accuracy and reliability for downstream analytics **by 15%**.
- Performed data wrangling on large, unstructured datasets, converting them into structured formats, which improved the efficiency of subsequent data analysis tasks **by 35%**.
- Leveraged Hive and HDFS for efficient data storage and querying, enabling quick access to historical data for trend analysis.

Tech Stack: PySpark, Apache Spark, Apache Kafka, Python, Apache Airflow, Tableau, K-means Clustering, Hive, HDFS, ETL, Data Wrangling, Data Pipelines.

PROJECTS:

YouTube Clone

<https://github.com/mhv2408/youtube-clone>

- Created a full stack in-house YouTube clone for storing, viewing videos using TypeScript, Next.js, Express.js, and Firebase.
- Implemented user authentication using Firebase Auth, facilitating secure Google account sign in/out for over 50 users.
- Created video upload functionality with Google Cloud Storage and signed URLs, enabling secure and efficient storage for large files up to 1 GB per video.
- Built an asynchronous video processing pipeline with Google Cloud Pub/Sub and Cloud Run, using FFmpeg for transcoding videos into multiple formats (360p, 720p) with an average processing time of 2 minutes per video.
- Initiated a responsive web client with Next.js, hosted on Google Cloud Run, allowing seamless viewing and navigation of a library containing over 100 videos.
- Established a robust backend with Firebase Functions to handle over 1,000 API requests per day, manage video metadata for 500+ videos using Firebase Firestore, and ensure seamless communication between frontend and backend.
- Employed Docker for containerization, achieving consistent development and deployment environments and reducing deployment time **by 30%**.

Tech Stack: TypeScript, Next.js, Express.js, Firebase Auth, GCS, Storage, Google Cloud Pub/Sub, Google Cloud Run, FFmpeg, Firebase Functions, Firebase Firestore, Docker.

SKILLS

Languages & Libraries: Python (Pandas, NumPy, SciPy), C,C++, C#, Java.

Web Technologies: HTML5, CSS3, JavaScript, TypeScript, ReactJs, Django, Flask, Node.js, Express.js., REST APIs, XML, JSON.

DevOps, Testing and CI/CD: Docker, Kubernetes, JUnit, JMeter, Maven, Jenkins.

Version Control: Git, GitHub, GitLab.

IDEs: Visual Studio Code, IntelliJ IDEA, Eclipse, Rstudio, Jupyter notebook.

Databases: Oracle, SQL Server, MySQL, MongoDB/NoSQL, PostgreSQL.

Cloud Technologies: AWS - S3, Lambda, SQS, SNS, RDS, EKS, EC2, IAM.

Operating Systems: Windows, Linux, MacOS.

Big Data Ecosystem: Hadoop, Apache Spark, HBase, MapReduce, HDFS, Hive, Sqoop, Data Bricks, Tableau, PowerBI.

Machine Learning: Supervised and Unsupervised Learning, Linear and Logistic Regression, Algorithms (Decision Trees, Ensemble models, Neural Networks, Deep Neural networks, RNN, CNN, SVM, Text Mining).

EDUCATION

Arizona State University, Tempe, AZ, USA
Master of Science, Computer Science

Jan 2023 - Dec 2024
CGPA: 4/4

Andhra University, Visakhapatnam, Andhra Pradesh, India
Bachelor of Technology, Computer Science and Engineering

Jun 2018 - May 2022
CGPA: 8.52/10